Round Robin Simulation:

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Course: CSCI330 - Operating Systems

Round Robin Explained:

A CPU-scheduling algorithm is a method operating systems use to select which process from the ready queue gets to be run on the CPU next. A preemptive scheduling algorithm goes one step further and decides when a process should be “preempted”. Preemption is when a process is “removed” or unloaded from the CPU and then sent back to the ready queue. The scheduler of the operating system enforces the algorithm by selecting the next process to be loaded and calls for the running process to be preempted. The scheduler’s commands are then sent to the dispatcher, which does the job of unloading and loading these processes to the CPU.

The Round Robin Algorithm is a preemptive scheduler algorithm. It uses a special variable, called a time Quantum, to determine the maximum cycle time a process is permitted to run on the CPU before being preempted to wait for its next turn.

Running Program Instructions (Windows Command Terminal):

Activating venv Python virtual environment:

This Python program uses a virtual environment to run its code.
Thus you must download python to begin running this program.
Run the following commands in the terminal in order.

1. If the venv folder is not present, run this to create a venv folder and it’s files:

python -m venv venv

2. Activating the environment:

python environment: venv\Scripts\Activate

3. To bypass security error you may encounter when trying to activate the previous script.

Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy Unrestricted -Force

Entering input via the commandline:

This “app” takes two arguments as input through the command line:

- file location to a CSV file (for processor data) and an integer value (Quantum time).
- It is advised to drop your CSV file in the “input” folder.
- If you enter no arguments or a bad argument.
    - It enters the default inputs:
    - **Input\sample2.csv** and **2**

WARNING: The current directory of your terminal when running the main script is important. If you dont run main in the correct directly the program will throw an error.

1. Open the command terminal within the “RoundRobinSimulation” folder:
    a. cd RoundRobinSimulation
2. Run the **main.py** file using the “python” command like so:
    a. Command Format:
    python main.py ..\input\ ***table_name.csv *single space**** (**integer number**)
    b. ***Examples****:*
        i. python main.py ..\input\sample2.csv 2
        ii. python main.py ..\input\sample3.csv 29
        iii. python main.py ..\input\sample1.csv 5

Simulation Code Summary:

My system depends on a time variable called "**timeCount**", which is initialized as zero, then incremented at the end of every simulation iteration.
Everything is executed within this single method of my Simulator class:
`def runSimulation(self, quantum : int )`
Every terminated process is added to a special "terminated" list.
When the length of the terminated list is equal to the number of processes entered into the simulation, the simulation stops.

The simulation class runs through the following steps each "tick" or iteration:

1. Check process arrival times of the current timeCount value.
    a. Receive the arriving Processes from ProcessHandler.
    b. Add the Process to the ready queue.
2. Run scheduler (tick)
    a. If the CPU is currently idle:
        i. Select the next process from the ready queue.
        ii. Load the new process to the CPU.
        iii. Reset the time quantum.
3. Run CPU
    a. Decrement the burst time of the loaded process.
4. Check if the running process has been terminated.
    a. IF terminated, unload the process and add it to the "terminated list."
5. If the process needs more processing time, decrement the time quantum.
6. Check if the time quantum has expired.
    a. IF the time quantum has expired, preempt the process, unload it from the CPU, and move it back to the ready queue.
    b. If the ready queue has another process available, load it into the CPU.
    c. If this newly loaded process has the same ID as the previous process, this means a CONTEXT SWITCH has occurred. And we increment the "switchCount" variable.
    d. Reset the time quantum.
7. END of iteration, return to step 1 until all processes are terminated.

Scripts and Classes:

Simulation.py Script:

Simulator class

- Responsible for running the simulation in a while loop.
- Prints simulation results when all processes terminate.
- **def runSimulation(self, quantum : int):**
    - Constructs Cpu, ProcessHandler, and RoundRobinScheduler objects.
    - Declares a *timeCount* variable, which is passed into the “tick” methods of said objects.
    - All the “tick” methods of each of these classes are called in this function, either directly or implicitly.

ProcessModule.py Script:

ProcessHandler class

- Stores references to all process objects in the simulation.
- **def receiveNextArrival(self,time : int) -> list[Process] | None:**
    - Uses a dictionary (Hashtable) to return a list of processes meant to arrive in the ready queue for the given “time” parameter.

processState Enum

- Enum States: NEW, READY, RUNNING, and TERMINATED
- A newly constructed process begins as NEW

Process State Transitions:

- NEW to READY:
    - When added to ReadyQueue by RoundRobinScheduler class using the method:
      def moveProcessToReadyQueue(self, p : Process):
- READY to RUNNING
    - Transition occurs within CPU class’s method:
      def loadProcess(self, p : Process)
- RUNNING to READY
    - Transition occurs within RoundRoblinSchedule class’s method:
      moveProcessToReadyQueue(self, p: Process).
- READY to TERMINATED
    - Transition occurs within Process class’s method:
      def decrementBurst(self,time : int) -> bool

Process class:

- Represents the processes being ran in our round robin simulation.
- Stores the total burst time (in totalBurst variable), which is how long a process needs to run in CPU to finish and terminate.
- Stores wait time and calculates turnaround time for itself.
- However, CPU is responsible for calling its burst time decrement method.

Constructor:

- **`def __init__( self, pid : int, arrive: int, totalBurst : int)`**

RoundRobinScheduler.py Script:

RoundRobinScheduler class

- Handles CPU object.
- Run's CPU tick method.

Cpu class:

- `def runProcess(self,time: int)`
  - Decrements the burst time of the process reference stored within its "runningProcess" variable.
  - Call's Process object's "decrementBurst" method.
- *`def unloadProcess(self) -> Process | None:`*
  - Sets runningProcess to None (Null), and sets CPU to IDLE state.
- *`def loadProcess(self, p : Process)`*
  - Passes the Process object reference to the runningProcess variable
  - Sets this new Process to its RUNNING state.
  - Sets CPU to RUNNING state.

CpuState Enum

- Enum States: RUNNING and IDLE

main.py Script:

Functions:

def readFileInput(fileLocation : str) -> list[tuple[int,...]] :

- Reads file location entered in the command line.
- Converts each row of numbers into a list of 3 element integer arrays.
  - Order: (pid, arrival time, burst time)

def parseCommandLineArgs() -> tuple[list[Process], int]:

- Extracts arguments from command line.
- Calls readFileInput method.
- Constructs process objects to pass to the simulator constructor.

Test Sample Outputs:

The process table from sample2.csv is based on the example problem from Professor Ravindrababu Ravula's video: https://www.youtube.com/watch?v=Sa600YsU16U

How to interpret output?:

Once the simulation is finished, it outputs a list of the terminated processes' data in the order they were terminated. Top first, bottom last. Each printed in the following format:

Process: ID: pid number AT: Arrival time BT: Burst time TAT: Turnaround time
WT: wait time CT: completion time

Afterwards, the overall CPU IDLE time, Total execution time, total switch count, Average Turnaround time, Average Waiting time, throughput, and CPU utilization.
This simulation keeps track of the number of context switches and assumes each context switch takes 1ms to perform.

Formulas Used:

Turnaround Time = Completion time – Arrival time

Average Turnaround Time = total sum of turnaround times / total number of processes

Average Waiting Time = total sum of waiting times / total number of processes

Throughput = total processes / total execution time

Idle time = 1 - (Context Switch time * number of context switches) / total execution time

CPU Utilization = (Total Execution Time – Idle Time) / Total Execution Time

Test input CSV (sample2.csv):

pid	arrive	burst
1	0	4
2	1	5
3	2	2
4	3	1
5	4	6
6	6	3

Simulation Outputs:

Quantum 2:

PROCESS TERMINATION METRICS:

Process: ID:3 AT:2 BT:2 TAT: 4 WT: 2 CT: 6
Process: ID:1 AT:0 BT:4 TAT: 8 WT: 4 CT: 8
Process: ID:4 AT:3 BT:1 TAT: 6 WT: 5 CT: 9
Process: ID:2 AT:1 BT:5 TAT: 17 WT: 12 CT: 18
Process: ID:6 AT:6 BT:3 TAT: 13 WT: 10 CT: 19
Process: ID:5 AT:4 BT:6 TAT: 17 WT: 11 CT: 21

CPU IDLE TIME: 0.48 ms
TOTAL EXECUTION TIME: 21 ms
TOTAL SWITCH COUNT: 11
Avg. TurnAroundTime: 10.83 ms
Avg. WaitingTime: 7.33 ms
Throughput: 0.29 ms
CPU Utilization: % 97.73

Quantum 3:

PROCESS TERMINATION METRICS:

Process: ID:3 AT:2 BT:2 TAT: 6 WT: 4 CT: 8
Process: ID:4 AT:3 BT:1 TAT: 6 WT: 5 CT: 9
Process: ID:1 AT:0 BT:4 TAT: 10 WT: 6 CT: 10
Process: ID:6 AT:6 BT:3 TAT: 10 WT: 7 CT: 16
Process: ID:2 AT:1 BT:5 TAT: 17 WT: 12 CT: 18
Process: ID:5 AT:4 BT:6 TAT: 17 WT: 11 CT: 21

CPU IDLE TIME: 0.62 ms
TOTAL EXECUTION TIME: 21 ms
TOTAL SWITCH COUNT: 8
Avg. TurnAroundTime: 11.0 ms
Avg. WaitingTime: 7.5 ms
Throughput: 0.29 ms
CPU Utilization: % 97.05

Quantum 4:

PROCESS TERMINATION METRICS:

Process: ID:1	AT:0	BT:4	TAT: 4	WT: 0 CT: 4
Process: ID:3	AT:2	BT:2	TAT: 8	WT: 6 CT: 10
Process: ID:4	AT:3	BT:1	TAT: 8	WT: 7 CT: 11
Process: ID:6	AT:6	BT:3	TAT: 12	WT: 9 CT: 18
Process: ID:2	AT:1	BT:5	TAT: 18	WT: 13 CT: 19
Process: ID:5	AT:4	BT:6	TAT: 17	WT: 11 CT: 21

CPU IDLE TIME: 0.67 ms
TOTAL EXECUTION TIME: 21 ms
TOTAL SWITCH COUNT: 7
Avg. TurnAroundTime: 11.17 ms
Avg. WaitingTime: 7.67 ms
Throughput: 0.29 ms
CPU Utilization: % 96.83

Quantum 5:

PROCESS TERMINATION METRICS:

Process: ID:1	AT:0	BT:4	TAT: 4	WT: 0 CT: 4
Process: ID:2	AT:1	BT:5	TAT: 8	WT: 3 CT: 9
Process: ID:3	AT:2	BT:2	TAT: 9	WT: 7 CT: 11
Process: ID:4	AT:3	BT:1	TAT: 9	WT: 8 CT: 12
Process: ID:6	AT:6	BT:3	TAT: 14	WT: 11 CT: 20
Process: ID:5	AT:4	BT:6	TAT: 17	WT: 11 CT: 21

CPU IDLE TIME: 0.71 ms
TOTAL EXECUTION TIME: 21 ms
TOTAL SWITCH COUNT: 6
Avg. TurnAroundTime: 10.17 ms
Avg. WaitingTime: 6.67 ms
Throughput: 0.29 ms
CPU Utilization: % 96.6

Quantum 6:

PROCESS TERMINATION METRICS:

Process: ID:1 AT:0 BT:4 TAT: 4 WT: 0 CT: 4
Process: ID:2 AT:1 BT:5 TAT: 8 WT: 3 CT: 9
Process: ID:3 AT:2 BT:2 TAT: 9 WT: 7 CT: 11
Process: ID:4 AT:3 BT:1 TAT: 9 WT: 8 CT: 12
Process: ID:5 AT:4 BT:6 TAT: 14 WT: 8 CT: 18
Process: ID:6 AT:6 BT:3 TAT: 15 WT: 12 CT: 21

CPU IDLE TIME: 0.76 ms
TOTAL EXECUTION TIME: 21 ms
TOTAL SWITCH COUNT: 5
Avg. TurnAroundTime: 9.83 ms
Avg. WaitingTime: 6.33 ms
Throughput: 0.29 ms
CPU Utilization: % 96.37

Output Analysis:

Criteria	Quantum 2	Quantum 3	Quantum 4	Quantum 5	Quantum 6
Avg. TAT	10.83	11.0	11.17	10.17	9.83
Avg. WT	7.33	7.5	7.67	6.67	6.33
Throughput	0.29	0.29	0.29	0.29	0.29
Cpu Utilization	97.73	96.83	96.83	96.6	96.37

We want to increase Throughput and CPU utilization.

While decreasing average turnaround time and average wait time.

So our goal should be to pick the time quantum that best fits these criteria.

Time quantum 2 has the highest CPU utilization percentage (97.73 %), but the average waiting time (10.83) and turnaround time (7.33).

Quantum 6 has the lowest average waiting time (6.33) and turnaround time (9.83).

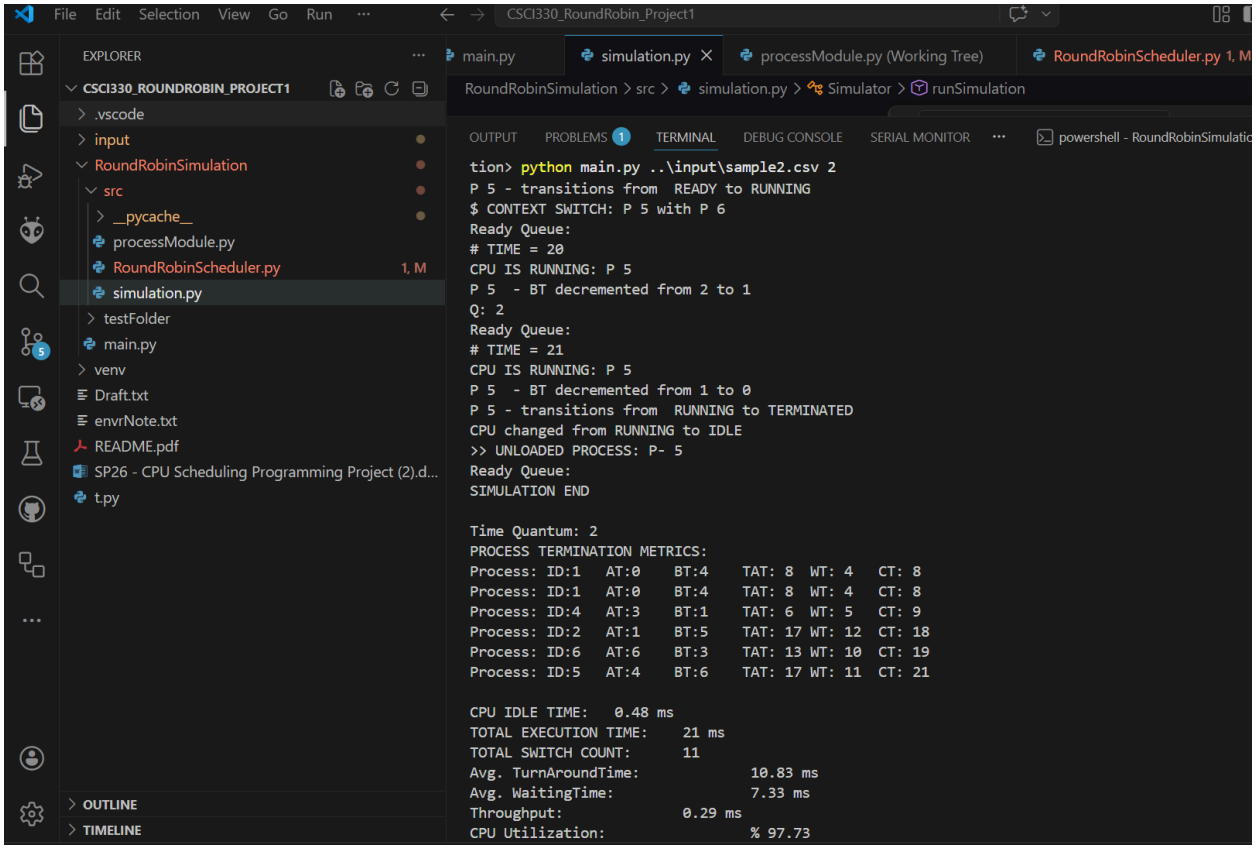
All time quantums resulted in the same throughput, 0.29, so we can ignore it to make our final decision.

In conclusion, If you value CPU utilization most for your operating system, you would choose time quantum 2. But if your priority is lower waiting times and turnaround time for your processes, you would chose quantum 6.

But because quantum 6 gives so much time to each running process, it effectively degrades into a non-preemptive First Come First Serve scheduler. Wherein each process is executed until completion in the order they arrive to the ready queue without any preemption.

Thus, choosing quantum 6 to handle processes with burst times less than 6 ms, completely undermines the round robin implementation.

Screenshots of simulation outputs:



```
Time Quantum: 3
PROCESS TERMINATION METRICS:
Process: ID:3   AT:2    BT:2    TAT: 6  WT: 4   CT: 8
Process: ID:4   AT:3    BT:1    TAT: 6  WT: 5   CT: 9
Process: ID:1   AT:0    BT:4    TAT: 10 WT: 6   CT: 10
Process: ID:6   AT:6    BT:3    TAT: 10 WT: 7   CT: 16
Process: ID:2   AT:1    BT:5    TAT: 17 WT: 12  CT: 18
Process: ID:5   AT:4    BT:6    TAT: 17 WT: 11  CT: 21

CPU IDLE TIME:   0.62 ms
TOTAL EXECUTION TIME:    21 ms
TOTAL SWITCH COUNT:      8
Avg. TurnAroundTime:             11.0 ms
Avg. WaitingTime:                7.5 ms
Throughput:              0.29 ms
CPU Utilization:                 % 97.05
```

```
CPU IS RUNNING: P 5
P 5  - BT decremented from 1 to 0
P 5 - transitions from  RUNNING to TERMINATED
CPU changed from RUNNING to IDLE
>> UNLOADED PROCESS: P- 5
Ready Queue:
SIMULATION END

Time Quantum: 4
PROCESS TERMINATION METRICS:
Process: ID:1   AT:0    BT:4    TAT: 4  WT: 0   CT: 4
Process: ID:3   AT:2    BT:2    TAT: 8  WT: 6   CT: 10
Process: ID:4   AT:3    BT:1    TAT: 8  WT: 7   CT: 11
Process: ID:6   AT:6    BT:3    TAT: 12 WT: 9   CT: 18
Process: ID:2   AT:1    BT:5    TAT: 18 WT: 13  CT: 19
Process: ID:5   AT:4    BT:6    TAT: 17 WT: 11  CT: 21

CPU IDLE TIME:   0.67 ms
TOTAL EXECUTION TIME:    21 ms
TOTAL SWITCH COUNT:      7
Avg. TurnAroundTime:             11.17 ms
Avg. WaitingTime:                7.67 ms
Throughput:              0.29 ms
CPU Utilization:                 % 96.83
```

```
Time Quantum: 5
PROCESS TERMINATION METRICS:
Process: ID:1   AT:0    BT:4    TAT: 4  WT: 0   CT: 4
Process: ID:2   AT:1    BT:5    TAT: 8  WT: 3   CT: 9
Process: ID:3   AT:2    BT:2    TAT: 9  WT: 7   CT: 11
Process: ID:4   AT:3    BT:1    TAT: 9  WT: 8   CT: 12
Process: ID:6   AT:6    BT:3    TAT: 14 WT: 11  CT: 20
Process: ID:5   AT:4    BT:6    TAT: 17 WT: 11  CT: 21

CPU IDLE TIME:   0.71 ms
TOTAL EXECUTION TIME:    21 ms
TOTAL SWITCH COUNT:      6
Avg. TurnAroundTime:             10.17 ms
Avg. WaitingTime:                6.67 ms
Throughput:              0.29 ms
CPU Utilization:                 % 96.6
```

```
CPU IS RUNNING: P 6
P 6  - BT decremented from 2 to 1
Q: 5
Ready Queue:
# TIME = 21
CPU IS RUNNING: P 6
P 6  - BT decremented from 1 to 0
P 6 - transitions from  RUNNING to TERMINATED
CPU changed from RUNNING to IDLE
>> UNLOADED PROCESS: P- 6
Ready Queue:
SIMULATION END

Time Quantum: 6
PROCESS TERMINATION METRICS:
Process: ID:1   AT:0    BT:4    TAT: 4  WT: 0   CT: 4
Process: ID:2   AT:1    BT:5    TAT: 8  WT: 3   CT: 9
Process: ID:3   AT:2    BT:2    TAT: 9  WT: 7   CT: 11
Process: ID:4   AT:3    BT:1    TAT: 9  WT: 8   CT: 12
Process: ID:5   AT:4    BT:6    TAT: 14 WT: 8   CT: 18
Process: ID:6   AT:6    BT:3    TAT: 15 WT: 12  CT: 21

CPU IDLE TIME:   0.76 ms
TOTAL EXECUTION TIME:    21 ms
TOTAL SWITCH COUNT:      5
Avg. TurnAroundTime:              9.83 ms
Avg. WaitingTime:                 6.33 ms
Throughput:              0.29 ms
CPU Utilization:                  % 96.37
```